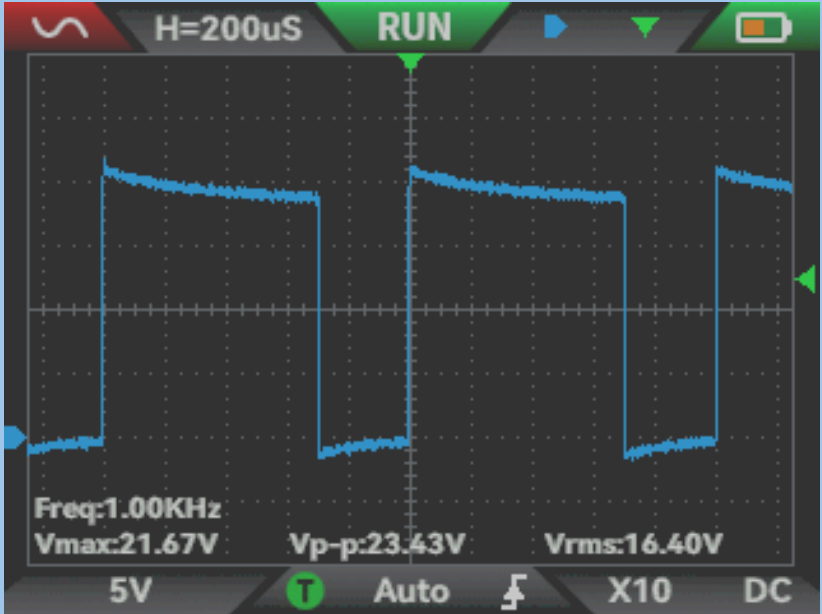


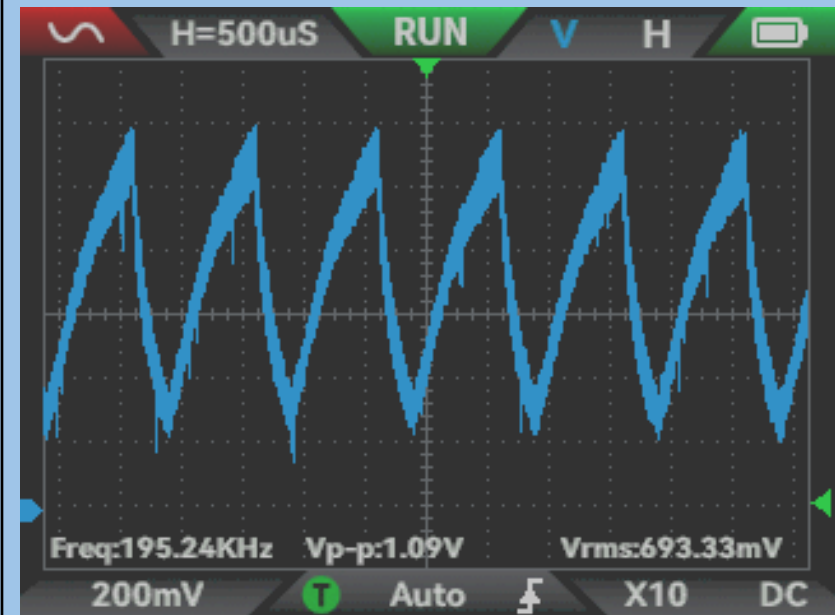
Notes	Image
Phase node 20.11v, .7 duty, 1khz, 550mA, 13bmp	

PTC fuse

20.11v, .7, 1kHz,

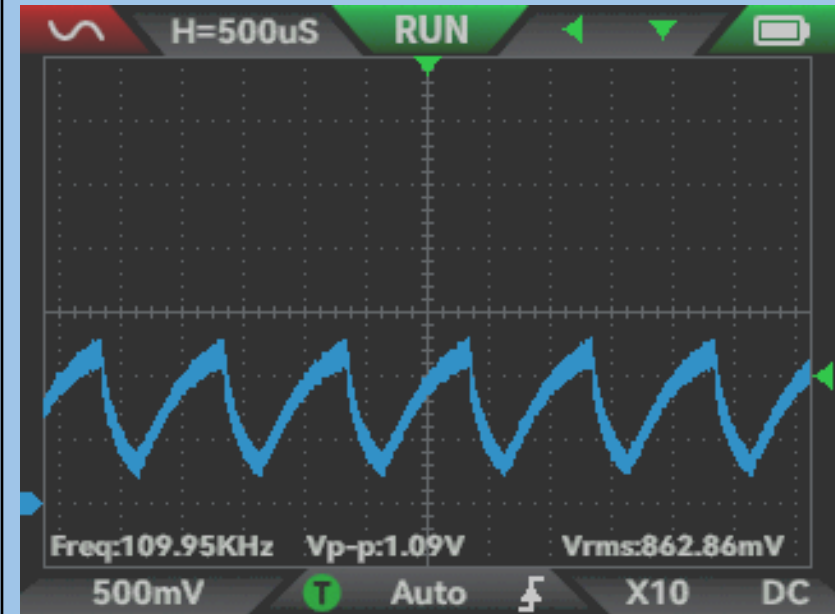
- 963mA (62mA idle) on total current

15bmp



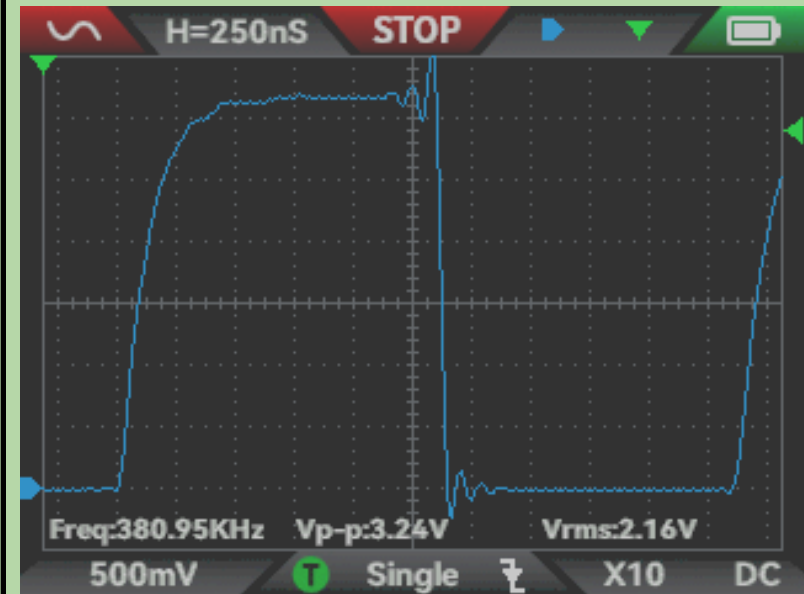
Phase current (remove extra current used by bbc)

- 23.75V,
- 70% duty
- 1kHz,
- 70.6mA (38mA no BBC) idle
- 3bmp, .67A



I2c at 400KHZ,

- 2.625us period (381.kHz)
- ~150ns to 2.02V ($1-1/e$)*3.18V
- ~260ns to 2.74V (2rc)
- 450ns to 3.16V
- 212-221us transmit time
 - 213us default

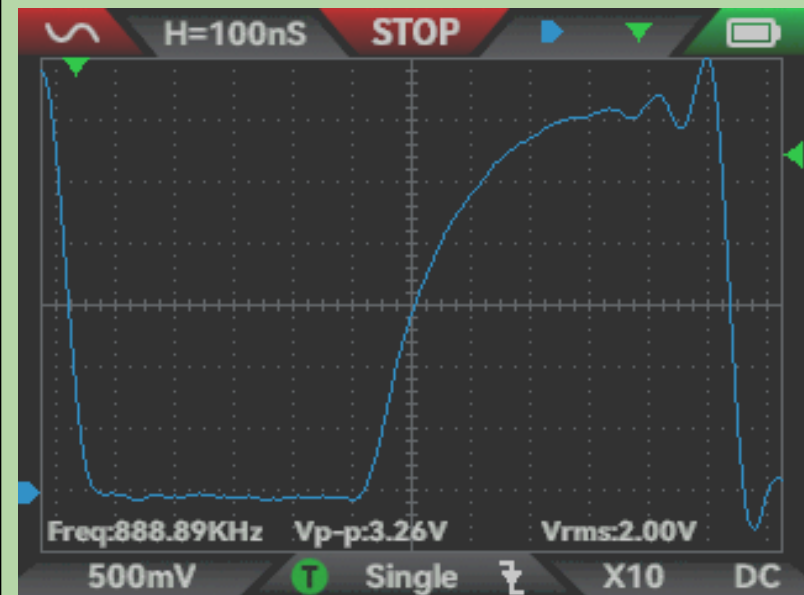


Same test at "1MHz" (shown)

- 110ns period (only around 900KHz)
- Slightly longer than 150ns to 2V
- I2c dies on touch
- 147-8 us

900kHz- 1.225us period

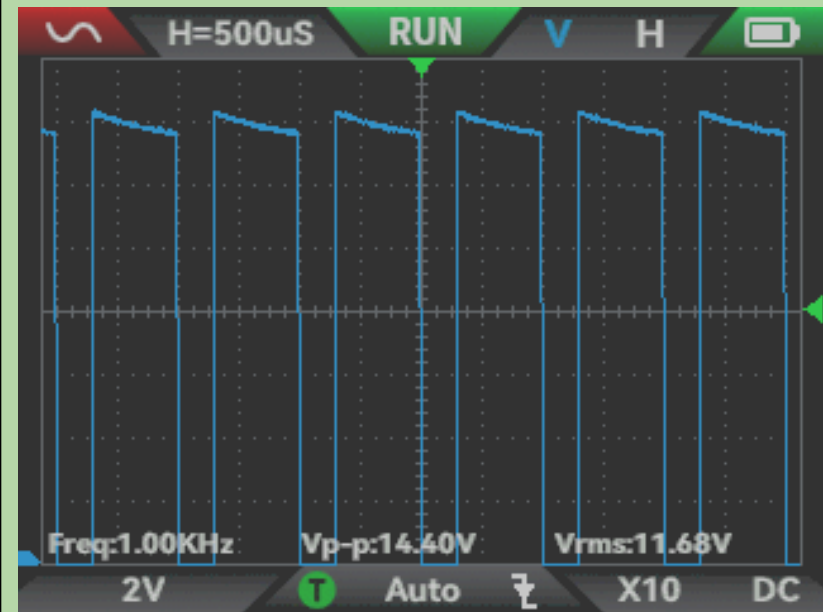
- 151-157us transmit time when touched
 - (Default 152us)
- Dies on hard big touch



Phase current b-c

12.62bat, 14.42 bbc, 1khz .7 duty, BHCL

- $V_{rms} = 11.68V$
- 417mA rms current, 52mA idle
- 5v from computer, Buck led on



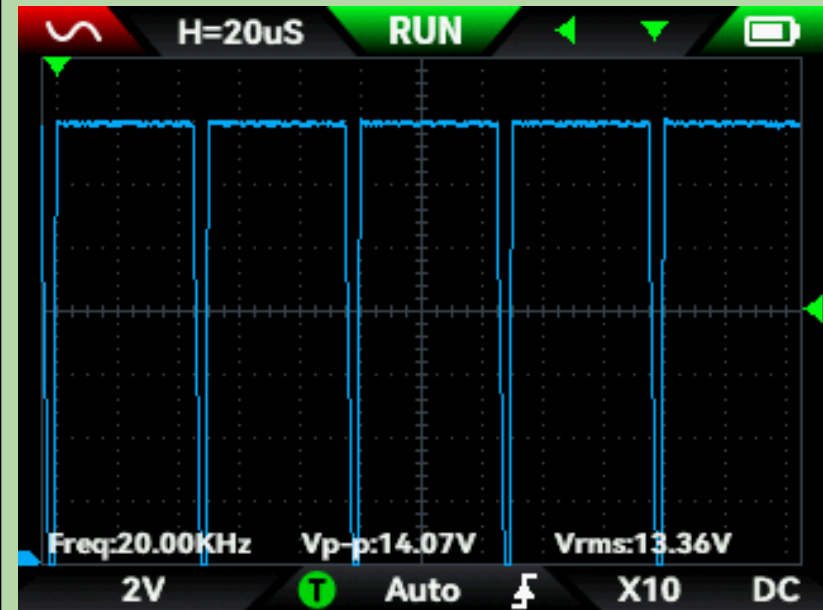
Phase current BHCL

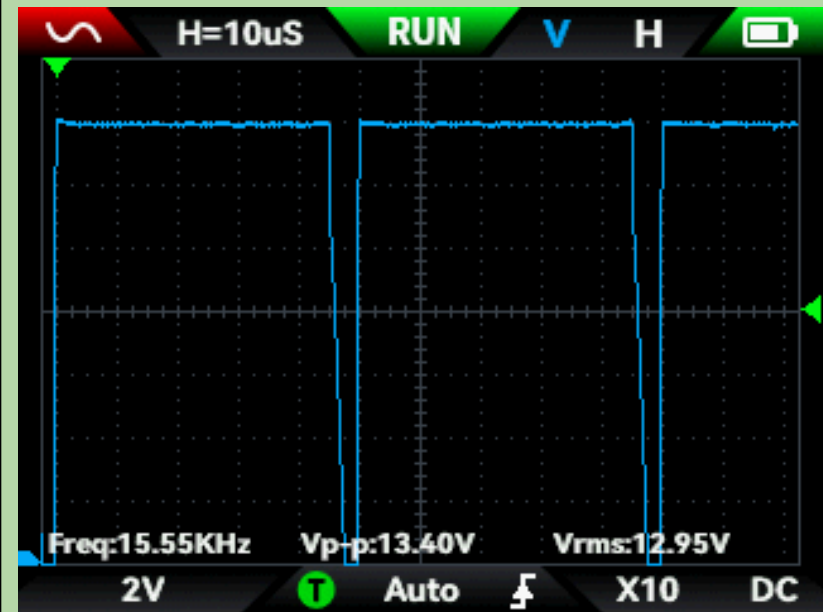
12.59bat, 14.42 bbc, 20khz 90% duty

- 50.2mA idle (I_T)
- $I_{rms} = 559mA$
- $I_{rms} = 557mA$ try 2 for high res

6/22/2025 day summary

- Added 5.1kHz in parallel to both i2c pullups
- Reoriented 5V 18awg line
- Soldered off a PTC and soldered 2 new ones in parallel in place
- Removed tvs2200 that was shorting and replaced it with a new one. Old one lost
- 330mA freespin, 720A at stall
- Achieve 1670RSP with 8000 tick as5600Pol period at clock of 4MHz.
- Working on Vtimer





6/25/2025

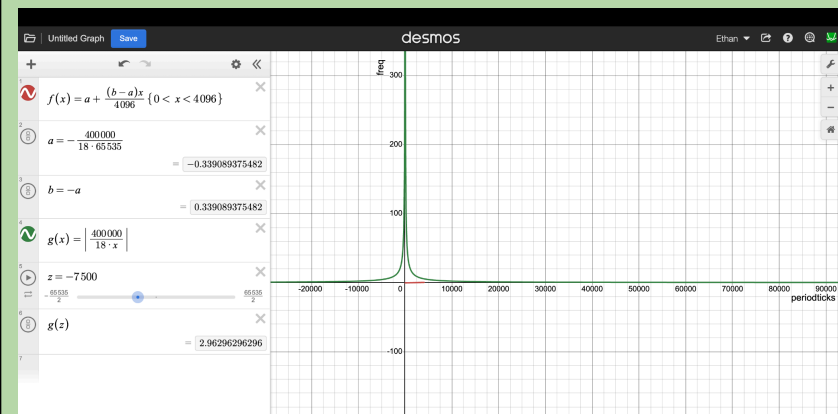
Ran motor at 23V with 95% duty cycle

- reached I_{rms} phase of 900mA +
- broke oscilloscope ish

Final (ok) working test

- 17.25V , 80% duty cycle
- 600mA rms Phase Current at stall
- ~50mA rms free spin current
- 430mA rms total freespining
- 880mA-1.09A stall (total)
- Tested and blew up Low Side TVS2200DRVR. Its the third time this happened, and it always happens on low side mosfets

Possible causes



- Overvoltage /overcurrent (from di/dt or dv/dt) - Backemf -	